

Reconstructing $f(R)$ gravity from generalized entropies: exact Lagrangians

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Generalized horizon entropies are widely used as theoretical modifications of the Bekenstein-Hawking area law, but through the Wald construction they may also encode modifications of the underlying gravitational dynamics. We reconstruct metric $f(R)$ gravity from prescribed entropy-area relations and show that the procedure is intrinsically branch dependent through the required area-curvature map. On the maximally symmetric branch, where $A = 48\pi/R$ exactly, the reconstruction reduces to a single quadrature and can be performed non-perturbatively. We obtain closed-form Lagrangians for several generalized entropies and show that an entropy term aS_{BH}^q generates a curvature term proportional to R^{2-q} . In particular, Kaniadakis entropy produces a $1/R$ correction, while logarithmic entropy generates an $R^2 \ln R$ term. We further derive a branch-independent criterion, $\partial_R^2 f = (ds/dR)d(S/s)/ds$, relating Dolgov-Kawasaki stability directly to the entropy functional, together with $m_{\text{sc}}^2 = S'(s)/(3f_{RR})$ on the maximally symmetric branch. Comparison with the fixed-mass Schwarzschild-de Sitter branch reveals different reconstructed Lagrangians and reversed stability properties. Finally, the weak-isolated-horizon boost charge reproduces the original generalized entropy. These results establish a direct non-perturbative link between generalized horizon thermodynamics and modified gravitational dynamics.

I. INTRODUCTION

The Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4G)$ provides one of the most fundamental connections between gravitation, thermodynamics and quantum theory [1, 2]. Nevertheless, there are several independent reasons why the entropy associated with a gravitational horizon may depart from the standard area law. Long-range interactions, non-extensive statistical behavior, relativistic generalizations of statistical mechanics, non-equilibrium effects, quantum correlations and possible modifications of the microscopic horizon structure may all lead to generalized entropy functionals. In particular, Tsallis [3, 4] and Rényi [5, 6] entropies arise from generalized statistical frameworks, Kaniadakis entropy [7, 8] is motivated by a relativistic extension of statistical mechanics, while Barrow entropy [9] is associated with a possible fractal deformation of the horizon induced by quantum-gravitational effects. Despite their different motivations, such entropies can be applied to gravitational horizons and typically recover the Bekenstein-Hawking form in appropriate limits [10–13]. Additionally, when applied

to cosmological horizons, they can lead to modified cosmological scenarios through the gravity-thermodynamics conjecture [14–31], or to extended holographic dark energy scenarios [32–56].

Since the horizon entropy is modified, it is natural to ask what gravitational dynamics correspond to this modification. For any diffeomorphism-invariant theory the entropy of a stationary horizon is determined by the associated Noether charge [57–59]. In metric $f(R)$ gravity [60, 61] the Wald entropy is $S_{\text{Wald}} = \frac{A}{4G} f_R$. Thus, a prescribed entropy-area relation determines f_R on the horizon and can in principle be used to reconstruct the gravitational Lagrangian. This inverse problem is also motivated by the broader relation between gravitational dynamics and horizon thermodynamics suggested by thermodynamic and emergent-gravity approaches [57–59] (see also [62–64]).

The main difficulty is that the entropy is naturally a function of the horizon area, whereas the Lagrangian is a function of the curvature. Hence, an area-curvature relation $A(R)$ must be specified. As we show, this relation is not unique. Schwarzschild-de Sitter constant-curvature geometries depend on both the curvature and the mass parameter, and therefore the horizon area is not uniquely fixed by R . The reconstruction becomes well defined only after a one-parameter branch in this space of geometries is selected. Different branches consequently lead to different entropy-Lagrangian correspondences.

We investigate two physically distinct choices. For a

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Schwarzschild-de Sitter black hole of fixed mass, the area-curvature relation is known through an expansion around the Schwarzschild solution. On the maximally symmetric branch, however, the cosmological horizon satisfies $A = \frac{48\pi}{R}$, which is exact and mass independent. In this case the Wald relation reduces to a single quadrature and the corresponding $f(R)$ Lagrangian can be reconstructed without expanding in the entropy deformation parameter.

Using this exact construction, we obtain closed-form Lagrangians for the Bekenstein-Hawking, Tsallis-Cirto, Barrow, Rényi, Kaniadakis, logarithmic and power-law entanglement entropies. We further derive a general mapping between entropy and curvature powers. An entropy contribution proportional to S_{BH}^q generates a curvature contribution proportional to R^{2-q} , with $q = 2$ yielding a logarithmic term. Hence, the exponent is reflected about the extensive value $q = 1$, directly determining the infrared or higher-curvature character of the induced modification. In particular, the Kaniadakis entropy generates at leading order the $1/R$ modification of Carroll, Duvvuri, Trodden and Turner [65], while the logarithmically corrected entropy produces the $R^2 \ln R$ structure appearing in one-loop effective descriptions [66–68].

Furthermore, we derive general viability conditions directly from the entropy functional. A branch-independent relation expresses f_{RR} through the monotonicity of $S(s)/s$, where $s = S_{\text{BH}}$, and therefore turns the Dolgov-Kawasaki condition into an entropic criterion whose sign is reversed between the two branches. On the maximally symmetric branch the scalaron mass can additionally be written directly in terms of $S'(s)$. Hence, ghost freedom, Dolgov-Kawasaki stability and the absence of a tachyonic scalaron can all be related to properties of the entropy itself.

For comparison, we analyze the massive branch and reconstruct the Rényi, Barrow and Kaniadakis models within its regime of validity, examining their constant-curvature vacua and scalaron stability. The comparison shows that the two branches are not different approximations to the same theory, but different reconstructions associated with different horizon families. Finally, we express the reconstructed theories in first-order scalar-tensor form and study them within the covariant phase space of weak isolated horizons [69–72]. The corresponding conserved boost charge reproduces the generalized entropy, providing an independent quasi-local consistency check.

The structure of the paper is as follows. In section II we present the reconstruction framework, the area-curvature ambiguity, the two branches and the exact maximally symmetric construction. In section III we derive the exact Lagrangians and examine their constant-curvature vacua, stability and infrared properties. In section IV we study the massive branch, compare the two reconstructions and derive the corresponding weak isolated-horizon charges. Finally, in section V we summarize

the results and discuss possible extensions. We use $G = c = \hbar = 1$ except where G is displayed for clarity, and write $\partial_R f \equiv f_R$ and $\partial_R^2 f \equiv f_{RR}$.

II. RECONSTRUCTION FRAMEWORK

In this section we formulate the reconstruction procedure that relates a prescribed horizon entropy to an $f(R)$ gravitational Lagrangian. We first identify the role of the area-curvature relation and its branch dependence, and then derive the corresponding viability conditions and the exact reconstruction on the maximally symmetric branch.

A. Master relation and area-curvature ambiguity

For the $f(R)$ action [60, 73–82]

$$I = \frac{1}{16\pi G} \int d^4x \sqrt{-g} f(R), \quad (1)$$

the Wald entropy of a stationary horizon is [60, 61]

$$S_{\text{Wald}} = \frac{A}{4G} \partial_R f, \quad (2)$$

which reduces to $A/4G$ for $f(R) = R - 2\Lambda$. Demanding that Eq. (2) is equal to a prescribed generalized entropy $S(A)$ gives the master relation of the reconstruction programme

$$\partial_R f = \frac{4G S(A)}{A}. \quad (3)$$

Since most generalized entropies recover S_{BH} as their corresponding deformation parameter vanishes, one may write the generalized entropy as an expansion in this deformation parameter according to

$$S = S_{\text{BH}} [1 + \epsilon \Phi(A) + \mathcal{O}(\epsilon^2)], \quad (4)$$

where ϵ is the deformation parameter and $\Phi(A)$ is a model-dependent function of the horizon area. In this language, the Einstein limit is approached continuously and ϵ acquires the interpretation of an effective coupling that measures departures from General Relativity. We shall use Eq. (4) in the analysis of the massive branch below. However, as we will show, the full Lagrangian can also be recovered in closed form without such a perturbative expansion if one assumes the existence of a maximally symmetric solution.

We mention here that relation (3) is exact. It determines $\partial_R f$ as a function of the horizon area, whereas the Lagrangian is a function of the Ricci scalar. Hence, the reconstruction is complete only once a relation $A = A(R)$ has been supplied. Let us therefore examine this relation in more detail.

We consider Schwarzschild-de Sitter geometries within metric $f(R)$ gravity, given by

$$ds^2 = -h(r)dt^2 + h(r)^{-1}dr^2 + r^2d\Omega^2, \quad (5)$$

$$h(r) = 1 - \frac{2Gm}{r} - \frac{Rr^2}{12},$$

whose horizon radius satisfies

$$\frac{R}{12}r_h^3 - r_h + 2Gm = 0. \quad (6)$$

The viability conditions for the existence of such solutions will be discussed in the following subsection. Since Eq. (6) may be solved for the mass as $2Gm = r_h - Rr_h^3/12$, for any chosen R and r_h it fixes a corresponding mass parameter m . Thus, at the level of the Schwarzschild-de Sitter ansatz, the horizon area is not uniquely determined by the scalar curvature only. Consequently, there is no unique canonical function $A(R)$, and Eq. (3) becomes well posed only after a one-parameter curve through this family specified by (M, Λ) has been selected. Different curves define different reconstructions, and this choice is the principal input of the method. We stress that this is not an approximation. Two natural choices are the massive branch and the maximally symmetric branch.

(i) *The massive branch.* The horizon radius can be found by solving Eq. (6) for r_h . Expanding around the Schwarzschild point $R = 0$, one obtains $r_h = 2GM(1 + G^2M^2R/3) + \mathcal{O}(R^2)$. Therefore, the horizon area can be approximated as

$$A(R) = A_0(1 + cR), \quad A_0 \equiv 16\pi G^2M^2, \quad c \equiv \frac{2G^2M^2}{3}. \quad (7)$$

This map underpins the existing reconstruction method and will be used in the analysis of the massive branch below. It is the origin of the constant c appearing there and it has two limitations. It retains a dependence on M , and hence the reconstructed Lagrangian is not universal. Moreover, it is accurate only to leading order in the curvature expansion around $R = 0$, and therefore any reconstruction based on this map is controlled within this curvature regime.

(ii) *The maximally symmetric branch.* Setting $m = 0$ instead leaves a one-parameter family of de Sitter geometries with $R = 12/L^2$ and a cosmological horizon of radius L , yielding

$$A(R) = \frac{48\pi}{R}. \quad (8)$$

Equation (8) is exact, involves no expansion, and carries no free mass parameter. It therefore defines a reconstruction that is exact in the deformation parameter, and it is this property that allows for the closed-form Lagrangians derived below.

Both branches are legitimate, and neither is preferred on thermodynamic grounds alone. Each imposes Eq. (3)

along a different curve of vacua and each answers a different physical question, namely what Lagrangian reproduces a given entropy for a black hole of fixed mass or for a cosmological horizon. They agree, as they must, when the deformation is switched off, with both returning $f(R) = R - 2\Lambda$. However, they differ in the structure of the induced corrections, as we will discuss in detail below.

B. Viability conditions

A reconstructed Lagrangian is admissible only if it supports a maximally symmetric vacuum and a healthy scalar sector. Varying Eq. (1) gives

$$f_R R_{\mu\nu} - \frac{1}{2}f g_{\mu\nu} + (g_{\mu\nu}\square - \nabla_\mu \nabla_\nu)f_R = 0, \quad (9)$$

whose trace, $Rf_R - 2f + 3\square f_R = 0$, reduces at constant curvature to the algebraic condition

$$R_0 f_R(R_0) - 2f(R_0) = 0. \quad (10)$$

Every real root of equation (10) supplies a Schwarzschild-(anti-)de Sitter background of the form (5). Depending on the sign of the root, expression (5) with $R \rightarrow R_0$ describes a Schwarzschild-de Sitter ($R_0 > 0$), Schwarzschild ($R_0 = 0$), or Schwarzschild-anti-de Sitter ($R_0 < 0$) geometry. Hence, the reconstructed theory possesses genuine black-hole solutions rather than only a maximally symmetric vacuum.

Absence of ghosts requires a positive effective Newton constant, $f_R(R_0) > 0$. Absence of the Dolgov-Kawasaki instability requires $f_{RR}(R_0) > 0$ [83, 84]. Additionally, the scalaron, namely the additional scalar mode of metric $f(R)$ gravity, must be non-tachyonic,

$$m_{\text{sc}}^2 = \frac{f_R(R_0) - R_0 f_{RR}(R_0)}{3f_{RR}(R_0)} > 0. \quad (11)$$

These are the standard viability criteria [85–87], and we apply them to every reconstruction below.

C. Exact reconstruction on the maximally symmetric branch

On the branch (8) the Bekenstein-Hawking entropy of the cosmological horizon is

$$s \equiv S_{\text{BH}} = \frac{A}{4G} = \frac{\sigma}{R}, \quad \sigma \equiv \frac{12\pi}{G}, \quad (12)$$

so that $R = \sigma/s$ and $dR = -\sigma s^{-2}ds$. Writing the generalized entropy as a function of s , Eq. (3) reads $\partial_R f = S(s)/s$, and integrating gives the exact master formula

$$f(R) = -\sigma \int^{\sigma/R} \frac{S(s')}{s'^3} ds' + c_1, \quad (13)$$

with c_1 fixed by the undeformed limit. Equation (13) is the central result of this paper. It contains no expansion of any kind. Given $S(s)$, a single elementary quadrature returns the Lagrangian.

Its consistency can be directly verified. Differentiating, one obtains

$$\partial_R f = -\sigma \frac{S(s)}{s^3} \frac{ds}{dR} = \frac{\sigma^2}{R^2} \frac{S(s)}{s^3} = \frac{S(s)}{s}, \quad (14)$$

which reproduces Eq. (3) identically. Moreover, for the undeformed entropy $S(s) = s$,

$$f(R) = -\sigma \int^{\sigma/R} \frac{ds'}{s'^2} + c_1 = \frac{\sigma}{s} + c_1 = R - 2\Lambda, \quad (15)$$

and thus Einstein gravity is recovered identically rather than asymptotically. This is a stronger statement than can be made on the massive branch, where the Einstein limit holds only to the order at which the map (7) is valid.

The exact master formula also allows for a general relation between the functional form of the entropy and that of the resulting Lagrangian. Suppose that the entropy admits a generalized power-series representation

$$S(s) = \sum_k a_k s^{q_k}, \quad (16)$$

with arbitrary real exponents q_k . Term-by-term integration of Eq. (13) gives

$$f(R) = \sum_k \frac{a_k \sigma^{q_k-1}}{2-q_k} R^{2-q_k} + c_1, \quad q_k \neq 2, \quad (17)$$

while the exceptional exponent $q_k = 2$ contributes $a_k \sigma \ln R$ in place of the corresponding power. Equation (17) may therefore be summarized through the following rule.

A term $a S_{\text{BH}}^q$ in the entropy generates a term $\propto R^{2-q}$ in the Lagrangian.

Hence, the exponent is reflected about $q = 1$. The extensive term $q = 1$ maps to the Einstein term R . More importantly, entropy contributions with $q > 1$ generate infrared curvature corrections with $2-q < 1$. This inversion is the characteristic feature of the maximally symmetric branch. On the massive branch, where $s \propto 1+cR$, the exponents are preserved, while here they are reflected. This follows directly from the inverse relation (12) between entropy and curvature on a de Sitter horizon, where a larger horizon corresponds to a smaller curvature.

III. EXACT LAGRANGIANS AND PHYSICAL PROPERTIES

In this section we apply the exact reconstruction derived above to the generalized entropies of interest. We

obtain the corresponding closed-form $f(R)$ Lagrangians and then investigate their constant-curvature vacua, stability conditions and infrared or higher-curvature behavior.

A. Exact Lagrangians

We now apply Eq. (13) to the entropies of interest. All results below have been verified by differentiation against Eq. (3), both symbolically and by numerical quadrature, and are collected in Table I.

Tsallis-Cirto [3, 4] and Barrow [9] entropies. For $S_T = \gamma s^\delta$ the relation (17) applies with a single term,

$$f_T(R) = \frac{\gamma \sigma^{\delta-1}}{2-\delta} R^{2-\delta} - 2\Lambda, \quad \delta \neq 2. \quad (18)$$

The Barrow entropy is the case $\gamma = 1$ and $\delta = 1 + \Delta/2$, giving the pure power law

$$f_B(R) = \frac{\sigma^{\Delta/2}}{1-\Delta/2} R^{1-\Delta/2} - 2\Lambda. \quad (19)$$

The fractal exponent appears directly as a shift of the power of the Ricci scalar, and Eq. (19) is exact for all $\Delta \in [0, 1]$, without any smallness assumption. The marginal case $\delta = 2$ gives instead a purely logarithmic Lagrangian,

$$f_T(R)|_{\delta=2} = \gamma \sigma \ln R - 2\Lambda. \quad (20)$$

Rényi entropy [5, 6]. For $S(s) = \lambda^{-1} \ln(1 + \lambda s)$ the integral in Eq. (13) is elementary,

$$f_{\mathcal{R}}(R) = \frac{R^2}{2\lambda\sigma} \ln\left(1 + \frac{\lambda\sigma}{R}\right) + \frac{R}{2} - \frac{\lambda\sigma}{2} \ln\left(\frac{R + \lambda\sigma}{\sigma}\right) - 2\Lambda. \quad (21)$$

It is worth noting that the Rényi reconstruction, which on the massive branch requires a dilogarithm, is elementary here. Expanding for small λ , one obtains

$$f_{\mathcal{R}}(R) = R - \frac{\lambda\sigma}{2} \ln R - 2\Lambda + \text{const} + \mathcal{O}(\lambda^2), \quad (22)$$

and hence the leading correction is a curvature logarithm.

Kaniadakis entropy [7, 8]. For $S(s) = \kappa^{-1} \sinh(\kappa s)$, two integrations by parts give

$$f_K(R) = \frac{R^2}{2\kappa\sigma} \sinh \frac{\kappa\sigma}{R} + \frac{R}{2} \cosh \frac{\kappa\sigma}{R} - \frac{\kappa\sigma}{2} \text{Shi}\left(\frac{\kappa\sigma}{R}\right) - 2\Lambda, \quad (23)$$

where $\text{Shi}(x) = \int_0^x \sinh(t) t^{-1} dt$ is the hyperbolic sine integral. Expanding for small κ , we find

$$f_K(R) = R - \frac{\kappa^2 \sigma^2}{6R} - 2\Lambda + \mathcal{O}(\kappa^4). \quad (24)$$

The leading Kaniadakis correction on the de Sitter branch is precisely the $1/R$ modification proposed by Carroll, Duvvuri, Trodden and Turner as a geometric origin of late-time acceleration [65], with the inverse-curvature scale fixed by the deformation parameter as $\kappa^2\sigma^2/6$. Its stability will be examined below.

Logarithmically corrected entropy. For $S(s) = s + \alpha \ln s$, characteristic of loop quantum gravity and microstate counting [66, 67], one obtains

$$f_{\log}(R) = R + \frac{\alpha R^2}{4\sigma} \left[1 + 2 \ln \frac{\sigma}{R} \right] - 2\Lambda . \quad (25)$$

The correction has the $R^2 \ln R$ form generated by the conformal anomaly and by one-loop effective actions in curved space [68]. Hence, a logarithmic entropy correction leads directly to a logarithmic curvature form factor, and this relation holds exactly in α rather than perturbatively.

Power-law entanglement entropy. For $S(s) = s(1 - \mu s^\beta)$ [88], Eq. (17) gives

$$f_{\text{pl}}(R) = R - \frac{\mu \sigma^\beta}{1 - \beta} R^{1-\beta} - 2\Lambda , \quad \beta \neq 1 . \quad (26)$$

B. Constant-curvature vacua

The reconstructed Lagrangians are physically admissible only if they possess maximally symmetric vacua. On this branch the trace condition (10) can be written in a general form. Setting $f(R) = F(s) - 2\Lambda$, with $F(s) \equiv -\sigma \int^s S(s') s'^{-3} ds'$, and using $\partial_R f = S(s)/s$ and $R = \sigma/s$, Eq. (10) becomes

$$\frac{\sigma S(s_0)}{s_0^2} - 2F(s_0) + 4\Lambda = 0 , \quad (27)$$

which is an algebraic equation for the horizon entropy $s_0 = \sigma/R_0$ of the vacuum. Equation (27) is exact and requires only the entropy functional and its quadrature. Applied to the single-power family (18), it gives

$$\frac{\gamma \delta \sigma^{\delta-1}}{2 - \delta} R_0^{2-\delta} = 4\Lambda , \quad (28)$$

and thus a real vacuum exists whenever $\Lambda(2 - \delta)/\delta > 0$, with

$$R_0 = \left[\frac{4\Lambda(2 - \delta)}{\gamma \delta \sigma^{\delta-1}} \right]^{\frac{1}{2-\delta}} . \quad (29)$$

Setting $\gamma = \delta = 1$ returns $R_0 = 4\Lambda$, namely the Einstein value, as required. For the Barrow entropy $\delta = 1 + \Delta/2 < 2$ throughout the physical range $\Delta \in [0, 1]$, and hence a real de Sitter vacuum exists for every admissible Barrow exponent. The same conclusion holds for the power-law entanglement entropy for $0 < \beta < 1$ and $\Lambda > 0$, for

which the constant-curvature equation admits at least one positive root.

For the Rényi and Kaniadakis reconstructions Eq. (27) is transcendental and must be solved numerically. Nevertheless, the existence of a root near the Einstein value follows from the implicit function theorem. In the zero-deformation limit the left-hand side of Eq. (27) reduces to $-\sigma/s_0 + 4\Lambda$, which vanishes at $s_0 = \sigma/(4\Lambda)$ and has a non-vanishing derivative there. Therefore, a root continuously connected to the Einstein solution exists for sufficiently small deformation.

Ghost freedom follows directly. Since $\partial_R f = S(s)/s$ and both S and s are positive, the effective Newton constant is automatically positive on this branch,

$$\partial_R f = \frac{S(s)}{s} > 0 , \quad (30)$$

for every entropy in Table I. Hence, the reconstruction does not introduce ghosts. The remaining viability conditions concern the sign of f_{RR} and the scalaron mass, which determine the Dolgov-Kawasaki and tachyonic stability, respectively. These conditions will be examined below.

C. Extensivity, stability and infrared structure

The exact reconstruction allows for a general statement about stability which holds for both branches. Differentiating the master relation (3), written as $\partial_R f = S(s)/s$, and using the chain rule, we obtain

$$\partial_R^2 f = \frac{ds}{dR} \frac{d}{ds} \left[\frac{S(s)}{s} \right] . \quad (31)$$

Equation (31) makes no reference to a specific branch, since the entire branch dependence lies in the sign of ds/dR . The Dolgov-Kawasaki condition $\partial_R^2 f > 0$ is therefore determined by the monotonicity of $S(s)/s$, namely by whether the generalized entropy is sub- or super-extensive, with the branch determining which behavior is required.

Maximally symmetric branch. In this case $s = \sigma/R$, and hence $ds/dR = -\sigma/R^2 < 0$. Therefore,

$$\frac{d}{ds} \left[\frac{S(s)}{s} \right] < 0 \iff \partial_R^2 f > 0 . \quad (32)$$

Thus, on the de Sitter branch the reconstructed theory is free of the Dolgov-Kawasaki instability if and only if the generalized entropy is sub-extensive.

On this branch the scalaron condition can also be expressed directly in terms of the entropy functional. Since $s = \sigma/R$, one has

$$R \partial_R^2 f = -s \frac{d}{ds} \left[\frac{S(s)}{s} \right] , \quad (33)$$

Entropy	$S(s)$, $s \equiv S_{\text{BH}} = \sigma/R$	Exact $f(R) + 2\Lambda$	Leading correction	DK stability
Bekenstein-Hawking	s	R	-	marginal
Tsallis-Cirto	γs^δ	$\frac{\gamma \sigma^{\delta-1}}{2-\delta} R^{2-\delta}$	$\propto R^{2-\delta}$	only if $\delta < 1$
Barrow	$s^{1+\Delta/2}$	$\frac{\sigma^{\Delta/2}}{1-\Delta/2} R^{1-\Delta/2}$	$\propto R^{1-\Delta/2}$	unstable ($\Delta > 0$)
Rényi	$\lambda^{-1} \ln(1 + \lambda s)$	Eq. (21), elementary	$-\frac{\lambda \sigma}{2} \ln R$	stable, all $\lambda > 0$
Kaniadakis	$\kappa^{-1} \sinh(\kappa s)$	Eq. (23), hyperbolic sine integral	$-\frac{\kappa^2 \sigma^2}{6} R^{-1}$	unstable
Logarithmic	$s + \alpha \ln s$	Eq. (25)	$\propto R^2 \ln R$	$\alpha(1 - \ln s) < 0$
Power-law entanglement	$s(1 - \mu s^\beta)$	$R - \frac{\mu \sigma^\beta}{1-\beta} R^{1-\beta}$	$\propto R^{1-\beta}$	stable if $\mu\beta > 0$

TABLE I. Exact $f(R)$ Lagrangians reconstructed from generalized horizon entropies on the maximally symmetric branch through the master formula (13), with $\sigma = 12\pi/G$. No expansion in the deformation parameter is involved. All entries have been verified by differentiation against Eq. (3) and by numerical quadrature. The last column follows from the sub-extensivity criterion (32).

and therefore

$$\partial_R f - R \partial_R^2 f = \frac{S(s)}{s} + s \frac{d}{ds} \left[\frac{S(s)}{s} \right] = \frac{dS}{ds}. \quad (34)$$

Using Eq. (11), we thus obtain

$$m_{\text{sc}}^2 = \frac{S'(s)}{3\partial_R^2 f}. \quad (35)$$

Hence, whenever the entropy is monotonically increasing, $S'(s) > 0$, the Dolgov-Kawasaki condition $\partial_R^2 f > 0$ automatically implies $m_{\text{sc}}^2 > 0$. Together with $\partial_R f = S(s)/s > 0$, the three viability requirements can therefore be expressed directly in terms of the entropy functional on the maximally symmetric branch.

Massive branch. Here $s = s_0(1 + cR)$, with $s_0 = A_0/4G$ and $c > 0$, and therefore $ds/dR > 0$. Hence, the inequality is reversed and Dolgov-Kawasaki stability requires super-extensivity, namely $d(S/s)/ds > 0$.

Let us now apply these criteria to the generalized entropies considered above on the maximally symmetric branch.

Rényi entropy. For $S/s = \ln(1 + \lambda s)/(\lambda s)$, one obtains

$$\frac{d}{ds} \left[\frac{S}{s} \right] = \frac{1}{\lambda s^2} \left[\frac{\lambda s}{1 + \lambda s} - \ln(1 + \lambda s) \right] < 0, \quad (36)$$

where the inequality follows from $x/(1+x) < \ln(1+x)$ for $x > 0$. Therefore, the Rényi reconstruction is Dolgov-Kawasaki stable for every $\lambda > 0$, without any further restriction on the deformation parameter. Moreover,

$$S'(s) = \frac{1}{1 + \lambda s} > 0, \quad (37)$$

and hence Eq. (35) shows that the scalaron is non-tachyonic as well.

Tsallis-Cirto and Barrow entropies. In this case $S/s = \gamma s^{\delta-1}$ and

$$\frac{d}{ds} \left[\frac{S}{s} \right] = \gamma(\delta - 1)s^{\delta-2}, \quad (38)$$

which is positive for $\delta > 1$. Hence, the reconstructed theory is unstable for every Barrow exponent $\Delta > 0$, while the sub-extensive Tsallis-Cirto regime $\delta < 1$ is Dolgov-Kawasaki stable. For $\gamma > 0$ and $\delta > 0$ one also has $S'(s) = \gamma \delta s^{\delta-1} > 0$. Therefore, the stable Tsallis-Cirto regime has a non-tachyonic scalaron, while the Barrow reconstruction has $m_{\text{sc}}^2 < 0$ whenever $\Delta > 0$.

Kaniadakis entropy. For $S/s = \sinh(\kappa s)/(\kappa s)$, we obtain

$$\frac{d}{ds} \left[\frac{S}{s} \right] = \frac{\kappa s \cosh(\kappa s) - \sinh(\kappa s)}{\kappa s^2} > 0, \quad (39)$$

for all $\kappa \neq 0$. Thus, the Kaniadakis reconstruction is Dolgov-Kawasaki unstable throughout. Since

$$S'(s) = \cosh(\kappa s) > 0, \quad (40)$$

Eq. (35) further implies $m_{\text{sc}}^2 < 0$ on this branch.

Logarithmic and power-law entanglement entropies. For $S = s + \alpha \ln s$, one has $S/s = 1 + \alpha s^{-1} \ln s$ and

$$\frac{d}{ds} \left[\frac{S}{s} \right] = \frac{\alpha(1 - \ln s)}{s^2}. \quad (41)$$

Therefore, Dolgov-Kawasaki stability requires $\alpha(1 - \ln s) < 0$. A negative logarithmic coefficient is stabilizing for horizons with $s < e$ and destabilizing for $s > e$, while the opposite behavior arises for $\alpha > 0$. Since realistic horizons have $s \gg 1$, the physically relevant condition is $\alpha > 0$. The scalaron condition additionally requires

$$S'(s) = 1 + \frac{\alpha}{s} > 0. \quad (42)$$

For $S = s(1 - \mu s^\beta)$, one finds

$$\frac{d}{ds} \left[\frac{S}{s} \right] = -\mu\beta s^{\beta-1}, \quad (43)$$

and therefore Dolgov-Kawasaki stability requires $\mu\beta > 0$. In addition, Eq. (35) requires

$$S'(s) = 1 - \mu(1 + \beta)s^\beta > 0 \quad (44)$$

for the scalaron to remain non-tachyonic.

The Kaniadakis case provides an independent check of the above criterion. Its leading Lagrangian (24) is the $1/R$ model proposed in [65], with $\mu^4 = \kappa^2 \sigma^2 / 6$. This model is known to suffer from the Dolgov-Kawasaki instability [83, 84]. Hence, the instability obtained directly from the super-extensive character of $\sinh(\kappa s)/(\kappa s)$ reproduces the known dynamical result without the need to analyze the field equations.

On the massive branch the sign in Eq. (31) is inverted. Thus, the Barrow and Kaniadakis reconstructions satisfy the Dolgov-Kawasaki condition, while the Rényi reconstruction does not. The explicit scalaron analysis below provides an independent check of this behavior.

The stability criterion is also closely related to the infrared structure of the exact Lagrangians. According to Eq. (17), an entropy term aS_{BH}^q generates a term proportional to R^{2-q} in the Lagrangian. Therefore, entropy contributions with $q > 1$ generate curvature terms with power smaller than unity and hence infrared modifications. However, the infrared or ultraviolet character is determined by the exponent q , whereas Dolgov-Kawasaki stability is determined by the sign of $d(S/s)/ds$. These two properties are therefore related but should not be identified.

This distinction is important when the coefficient of the entropy correction is negative. For example, the power-law entanglement correction scales with an exponent larger than unity and therefore generates the infrared term $R^{1-\beta}$. Nevertheless, for $\mu\beta > 0$ it satisfies $d(S/s)/ds < 0$ and is Dolgov-Kawasaki stable on the maximally symmetric branch.

The exponent mapping determines where in the curvature regime the reconstructed theories differ from General Relativity. In particular, the Barrow Lagrangian $R^{1-\Delta/2}$ has an exponent below unity, and its ratio to the Einstein term grows without bound as $R \rightarrow 0$ and tends to zero as $R \rightarrow \infty$. Similarly, the Kaniadakis correction $-\kappa^2 \sigma^2 / (6R)$ diverges as $R \rightarrow 0$, while the leading Rényi correction $-\lambda \sigma \ln R / 2$ becomes relevant at small curvature.

On the other hand, the logarithmic entropy correction produces the $R^2 \ln R$ structure of Eq. (25), which has an ultraviolet character. This case should be distinguished from the exceptional power $q = 2$ in Eq. (17). A term proportional to s^2 in the entropy generates a logarithmic term $\ln R$ in the Lagrangian, whereas a term proportional to $\ln s$ in the entropy generates the $R^2 \ln R$ structure.

This behavior follows from the inverse area-curvature relation on the maximally symmetric branch. Since a

large de Sitter horizon corresponds to small curvature according to Eq. (12), entropy contributions with powers larger than the area term are mapped to curvature contributions that become important in the infrared. Thus, entropy corrections motivated by quantum-gravitational considerations may appear on this branch as infrared modifications of the gravitational Lagrangian.

The Kaniadakis case makes this connection particularly clear. The generated $1/R$ term can drive late-time acceleration without a cosmological constant, with its scale determined by the entropy deformation parameter rather than introduced independently. At the same time, its known Dolgov-Kawasaki instability is reproduced directly by the entropic criterion, while Eq. (35) shows that the corresponding scalaron is tachyonic. Hence, the thermodynamic and dynamical conclusions are consistent.

The same picture applies to the remaining reconstructions. Barrow and Tsallis-Cirto entropies with $\delta > 1$ generate infrared modifications and are unstable on this branch. The power-law entanglement entropy generates the milder correction $R^{1-\beta}$ and satisfies the Dolgov-Kawasaki condition for $\mu\beta > 0$, provided that the additional scalaron condition $S'(s) > 0$ is also fulfilled. The Rényi reconstruction is sub-extensive and provides an infrared modification while satisfying both the Dolgov-Kawasaki and scalaron stability conditions for arbitrary positive λ . Whether it also satisfies the local-gravity and matter-era constraints required for a fully viable infrared modification is a separate issue, requiring the matter sector and the analysis of the scalaron on a cosmological background. This investigation lies beyond the scope of the present work.

IV. MASSIVE BRANCH AND HORIZON CHARGES

In this section we turn to the fixed-mass branch and reconstruct the corresponding $f(R)$ models within the regime of validity of the area-curvature expansion. We then examine their constant-curvature vacua and scalaron stability, compare them with the exact maximally symmetric reconstruction, and finally study the associated weak isolated-horizon charges.

A. Reconstructed Lagrangians and constant-curvature vacua

We now turn to the branch (7), which is the relevant one for a black hole of fixed mass. As discussed above, the corresponding area-curvature map is known perturbatively through an expansion around $R = 0$. Independently, in the analysis below we also treat the generalized entropy perturbatively in its deformation parameter. Throughout this section we use $G = 1$ in the massive-branch expressions, so that $s = S_{\text{BH}} = 6\pi c(1 + cR)$.

Each generalized entropy is written as

$$S = S_{\text{BH}} [1 + \epsilon \Phi(A)] + \mathcal{O}(\epsilon^2), \quad (45)$$

where ϵ is the corresponding deformation parameter. Hence, the Lagrangian can be written as

$$f(R) = R - 2\Lambda + \epsilon h(R) + \mathcal{O}(\epsilon^2). \quad (46)$$

The closed expressions presented below are resummed forms constructed to reproduce the corresponding perturbative corrections at the retained order in the deformation parameter. They should not be identified with the exact finite-deformation quadratures within the massive-branch map, which are presented in Appendix A.

Rényi case. From $S_{\mathcal{R}} = \lambda^{-1} \ln(1 + \lambda S_{\text{BH}}) = S_{\text{BH}} - \frac{\lambda}{2} S_{\text{BH}}^2 + \mathcal{O}(\lambda^2)$ one reads $\Phi_{\mathcal{R}} = -A/8$, and Eq. (3) integrates to

$$f_{\mathcal{R}}(R) = \frac{-2c\Lambda + (1 + cR)e^{-\frac{3}{2}\pi c\lambda(1+cR)} - 1}{c}, \quad (47)$$

whose expansion gives

$$h_{\mathcal{R}}(R) = -\frac{3\pi}{2}(1 + cR)^2. \quad (48)$$

The logarithmic deformation of the entropy translates into a quadratic curvature correction. We stress that Eq. (47) is a resummation which is faithful to first order in λ . The exact massive-branch Rényi reconstruction involves a dilogarithm, and statements at large λ based on Eq. (47) should therefore be understood accordingly.

Barrow case. From $S_B = S_{\text{BH}}^{1+\Delta/2} = S_{\text{BH}}[1 + \frac{\Delta}{2} \ln S_{\text{BH}}] + \mathcal{O}(\Delta^2)$ one reads $\Phi_B = \frac{1}{2} \ln(A/4)$ and obtains

$$f_B(R) = \frac{[2 + \Delta(\ln(6\pi c) - 1)](1 + cR)^{1+\frac{\Delta}{2}}}{2c} - \frac{1 + 2c\Lambda}{c}, \quad (49)$$

with

$$h_B(R) = \frac{1 + cR}{2c} [\ln(6\pi c(1 + cR)) - 1]. \quad (50)$$

Kaniadakis case. From $S_K = \kappa^{-1} \sinh(\kappa S_{\text{BH}}) = S_{\text{BH}} + \frac{\kappa^2}{6} S_{\text{BH}}^3 + \mathcal{O}(\kappa^4)$ one reads $\Phi_K = A^2/96$. The absence of a linear term follows from the symmetry $\kappa \rightarrow -\kappa$. We then obtain

$$f_K(R) = \frac{\sinh[2\sqrt{3}\pi c\kappa(1 + cR)]}{2\sqrt{3}\pi c^2\kappa} + c_1, \quad (51)$$

with $c_1 = -2\Lambda - 1/c$ fixed by the Einstein limit and

$$h_K(R) = 2\pi^2 c(1 + cR)^3. \quad (52)$$

Thus, on this branch the Rényi, Barrow and Kaniadakis entropies generate quadratic, logarithmic and cubic curvature corrections, respectively. The entropy exponents are preserved rather than reflected, in contrast to the maximally symmetric branch discussed above.

Having obtained the reconstructed Lagrangians, we proceed to examine their constant-curvature vacua.

Rényi case. Writing $\mathcal{A} = \lambda\pi GM^2$, Eq. (47) gives $f = R - 2\Lambda - \mathcal{A}(1 + cR)^2/c$ and $f_R = 1 - 2\mathcal{A}(1 + cR)$ to first order. The quadratic terms in Eq. (10) cancel identically, leaving the linear equation $R(2\mathcal{A} - 1) + 4\Lambda + 2\mathcal{A}/c = 0$, with solution $R = (4\Lambda + 2\mathcal{A}/c)/(1 - 2\mathcal{A})$. Hence,

$$R_0 = 4\Lambda + 8\pi GM^2 \Lambda \lambda + \frac{3\pi}{G} \lambda + \mathcal{O}(\lambda^2). \quad (53)$$

Barrow case. In this case the logarithms prevent Eq. (10) from reducing to a polynomial, and we solve perturbatively by setting $R_0 = 4\Lambda + \Delta R_1$. Evaluating the $\mathcal{O}(\Delta)$ contribution at $R = 4\Lambda$ and defining $X = 1 + 4c\Lambda$, we obtain

$$R_1 = 2\Lambda \ln(6\pi cX) - \frac{X}{c} [\ln(6\pi cX) - 1]. \quad (54)$$

Kaniadakis case. With $\alpha = 4\pi GM^2 \kappa/\sqrt{3}$, Eq. (10) becomes the cubic

$$\frac{\alpha^2 c^2}{6} R^3 - \left(1 + \frac{\alpha^2}{2}\right) R + \left(4\Lambda - \frac{\alpha^2}{3c}\right) = 0, \quad (55)$$

whose Cardano discriminant is

$$\Delta_c = \frac{108}{\alpha^6 c^6} [8 + \alpha^2(12 - 144\Lambda^2 c^2) + \alpha^4(24\Lambda c + 6)]. \quad (56)$$

For small α the discriminant is positive, and hence three distinct real roots exist. The existence of a root continuously connected to the Einstein branch follows from the implicit function theorem applied to

$$F(R, \alpha) = -R + 4\Lambda + \alpha^2(1 + cR)^2 \left(\frac{R}{6} - \frac{1}{3c}\right). \quad (57)$$

Since $F(4\Lambda, 0) = 0$ and $\partial_R F|_{(4\Lambda, 0)} = -1 \neq 0$, a unique differentiable root $R(\alpha)$ exists near $\alpha = 0$, with $R(0) = 4\Lambda$.

In all three cases a real maximally symmetric vacuum exists and is continuously connected to the Einstein solution. Therefore, each reconstructed theory admits Schwarzschild-(anti-)de Sitter black-hole solutions.

B. Scalaron stability and comparison of the two branches

We now evaluate Eq. (11) on the backgrounds obtained above. The results, summarized in Table II, differ significantly between the models.

Rényi case. From Eq. (47), we obtain

$$m_{\text{sc}}^2 = \frac{4 + 3c\pi\lambda[-2 + cR_0(2 - 3c\pi(1 + cR_0)\lambda)]}{9c^2\pi\lambda[-4 + 3c\pi(1 + cR_0)\lambda]}. \quad (58)$$

The denominator vanishes at $\lambda_p = \frac{4}{3\pi c(1 + cR_0)}$, which is also the point where f_{RR} changes sign. In particular, $f_{RR} < 0$ for $0 < \lambda < \lambda_p$ and $f_{RR} > 0$ for $\lambda > \lambda_p$.

The numerator of Eq. (58) can be written as

$$4 + 6c\pi(cR_0 - 1)\lambda - 9c^3\pi^2 R_0(1 + cR_0)\lambda^2, \quad (59)$$

and is therefore a downward parabola in λ . Its discriminant is

$$36c^2\pi^2 [(cR_0 - 1)^2 + 4cR_0(1 + cR_0)] > 0, \quad (60)$$

and the physically relevant positive root is

$$\lambda_0 = \frac{(cR_0 - 1) + \sqrt{(cR_0 - 1)^2 + 4cR_0(1 + cR_0)}}{3\pi c^2 R_0(1 + cR_0)}. \quad (61)$$

For $c > 0$ and $R_0 > 0$ one finds

$$\lambda_0 < \lambda_p. \quad (62)$$

Indeed, defining $x = cR_0 > 0$, one has

$$\frac{\lambda_0}{\lambda_p} = \frac{x - 1 + \sqrt{5x^2 + 2x + 1}}{4x} < 1, \quad (63)$$

where the inequality follows from $(3x + 1)^2 - (5x^2 + 2x + 1) = 4x(x + 1) > 0$ for $x > 0$.

The parameter space therefore separates into three regions.

- I. For $0 < \lambda < \lambda_0$, the numerator is positive while the denominator is negative. Hence $m_{sc}^2 < 0$, and the scalaron is tachyonic. Moreover, $f_{RR} < 0$.
- II. For $\lambda_0 < \lambda < \lambda_p$, both the numerator and the denominator are negative. Hence $m_{sc}^2 > 0$, and the scalaron is non-tachyonic. However, $f_{RR} < 0$, and therefore the Dolgov-Kawasaki condition is violated.
- III. For $\lambda > \lambda_p$, the numerator remains negative while the denominator becomes positive. Hence $m_{sc}^2 < 0$, although $f_{RR} > 0$.

Thus, the scalaron is tachyon-free only in the finite interval

$$\lambda_0 < \lambda < \lambda_p, \quad (64)$$

but this interval lies entirely in the region where $f_{RR} < 0$. Therefore, there is no positive- λ interval for which the resummed Rényi reconstruction simultaneously satisfies the Dolgov-Kawasaki condition and the non-tachyonic scalaron condition.

This result is consistent with the branch-independent relation (31). Since the Rényi entropy is sub-extensive, while the massive branch requires super-extensivity, the exact massive-branch reconstruction must have $f_{RR} < 0$. The sign change at λ_p in the resummed Lagrangian (47) occurs outside the regime in which the first-order reconstruction is controlled and should not be interpreted as a physical recovery of stability. The exact quadrature of Appendix A confirms that the Rényi reconstruction is Dolgov-Kawasaki unstable for every $\lambda > 0$.

Barrow case. Equation (49) gives

$$m_{sc}^2 = \frac{2 + cR_0(2 - \Delta)}{3c\Delta}, \quad (65)$$

whose numerator and denominator are positive for $c > 0$, $R_0 > 0$ and $0 < \Delta \leq 1$. The reconstructed Barrow theory is therefore stable over its entire physical parameter range, without any additional constraint on Δ . As $\Delta \rightarrow 0^+$,

$$m_{sc}^2 \sim \frac{2(1 + cR_0)}{3c\Delta} \rightarrow +\infty, \quad (66)$$

and thus the scalaron becomes infinitely massive and decouples. At the same time, the Barrow entropy continuously returns to S_{BH} and the theory returns to General Relativity, where the extra scalar mode is absent. At maximal deformation $\Delta = 1$, one still has

$$m_{sc}^2 = \frac{2 + cR_0}{3c} > 0, \quad (67)$$

and hence even the strongest admissible fractal deformation leaves the constant-curvature vacuum tachyon-free. Among the three models, this is the most robust behavior on the massive branch.

Kaniadakis case. Equation (51) gives

$$m_{sc}^2 = \frac{\sqrt{3}}{18c^2\pi\kappa} \coth\left(2\sqrt{3}\pi c(1 + cR_0)\kappa\right) - \frac{R_0}{3}, \quad (68)$$

whose sign cannot be determined in closed form. Expanding consistently in the perturbative regime $\kappa \ll 1$, we obtain

$$m_{sc}^2 = \frac{1}{36c^3\pi^2\kappa^2(1 + cR_0)} - \frac{2cR_0 - 1}{9c} - \frac{4\pi^2 c}{45}\kappa^2(1 + cR_0)^3 + \mathcal{O}(\kappa^4). \quad (69)$$

There is no linear term, in agreement with the symmetry $\kappa \rightarrow -\kappa$ of the Kaniadakis entropy. The leading contribution $\sim [36\pi^2 c^3(1 + cR_0)\kappa^2]^{-1} > 0$ diverges as $\kappa \rightarrow 0$, while the remaining terms remain finite. Therefore, the reconstructed Kaniadakis theory is free of tachyonic instabilities for sufficiently small κ . Moreover, Eq. (31) implies $f_{RR} > 0$ on the massive branch, since the Kaniadakis entropy is super-extensive. Beyond the perturbative regime, the competition between the hyperbolic cotangent and the $-R_0/3$ term may drive m_{sc}^2 through zero, but this regime lies outside the validity of the perturbative reconstruction.

The above results allow us to compare directly the two reconstruction branches. The two sets of Lagrangians are not approximations of one another. They correspond to different curves in the space of constant-curvature vacua and answer different physical questions. The massive branch determines the Lagrangian that reproduces a given entropy for a black hole of fixed mass, leading to corrections organized in powers of $1 + cR$ and retaining

a dependence on M . On the other hand, the maximally symmetric branch determines the Lagrangian associated with a cosmological horizon, leading to corrections organized in powers of R and containing no free parameter beyond $\sigma = 12\pi/G$.

The main structural difference between the two branches lies in the exponent mapping. On the massive branch $s \propto 1 + cR$, and a term $a s^q$ in the entropy generates a term proportional to $(1 + cR)^q$ in the Lagrangian. Hence, the exponent is preserved. On the maximally symmetric branch $s \propto R^{-1}$, and the same entropy term generates R^{2-q} , so the exponent is reflected about unity. For instance, the Barrow entropy generates a correction proportional to $(1 + cR)^{1+\Delta/2}$ on the massive branch and an infrared correction $R^{1-\Delta/2}$ on the maximally symmetric branch. These different forms arise because a different horizon is used in each reconstruction.

The Dolgov-Kawasaki stability properties are correspondingly inverted. Since the sign of ds/dR differs between the two branches, Eq. (31) implies that the same monotonicity property of $S(s)/s$ gives opposite conditions. Thus, the Barrow reconstruction is stable on the massive branch and unstable on the maximally symmetric one. The Kaniadakis reconstruction is perturbatively stable on the massive branch and unstable on the maximally symmetric one. Conversely, the Rényi reconstruction is Dolgov-Kawasaki stable on the maximally symmetric branch but unstable on the massive branch. The explicit scalaron analysis above is consistent with this result, since the only interval where the resummed Rényi scalaron is non-tachyonic lies entirely in the region $f_{RR} < 0$. Hence, there is no contradiction between the two branches, since they lead to different reconstructed Lagrangians.

The comparison shows that the entropy-Lagrangian correspondence depends on the branch on which the reconstruction is performed. Therefore, physical conclusions concerning the resulting gravitational theory should always be associated with the corresponding horizon branch. For cosmological applications, where the relevant horizon is cosmological, the exact maximally symmetric reconstruction is the natural choice. On the other hand, for the thermodynamics of an individual black hole, the massive branch is the appropriate one.

Entropy	$m_{sc}^2 > 0$	DK condition	Verdict
Rényi	$\lambda_0 < \lambda < \lambda_p$	$f_{RR} < 0$ in this interval	no viable window
Barrow	$0 < \Delta \leq 1$	$f_{RR} > 0$	stable
Kaniadakis	$\kappa \ll 1$	$f_{RR} > 0$	perturbatively stable

TABLE II. Scalaron and Dolgov-Kawasaki stability of the massive-branch reconstructions. For the Rényi case the scalaron is non-tachyonic only for $\lambda_0 < \lambda < \lambda_p$, but $f_{RR} < 0$ throughout this interval, and therefore no fully viable stability window exists. The Barrow and Kaniadakis results agree with the super-extensivity criterion (31).

C. First-order form and horizon charges

We finally verify, within the covariant phase space of weak isolated horizons, that the conserved charge of the reconstructed theory reproduces the entropy from which it was constructed. This provides an independent consistency check. The Wald entropy is defined at a Killing horizon of a stationary spacetime, whereas the isolated-horizon charge is quasi-local and does not require the existence of a global Killing field. The scalar-tensor representation is particularly useful here, since it makes the additional scalar degree of freedom explicit and provides the natural first-order formulation for the subsequent horizon-charge analysis.

Metric $f(R)$ gravity is dynamically equivalent to a non-minimally coupled scalar-tensor theory [85]. Introducing an auxiliary field χ , we write $\mathcal{A}[g, \chi] = (16\pi G)^{-1} \int \sqrt{-g} [f(\chi) - f_\chi(\chi)(\chi - R)]$, where $f_\chi = \partial_\chi f$. Variation with respect to χ enforces $\chi = R$ whenever $f_{\chi\chi} \neq 0$. Defining

$$\phi \equiv f_R(\chi), \quad V(\phi) = \phi\chi(\phi) - f(\chi(\phi)), \quad (70)$$

the action takes the Jordan-frame form $\mathcal{A} = (16\pi G)^{-1} \int \sqrt{-g} [\phi R - V(\phi)]$. In first-order variables it becomes

$$16\pi G \mathcal{A}(e, A) = \int_M \phi \Sigma_{IJ} \wedge F^{IJ} - \frac{1}{4!} \int_M V(\phi) \epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L, \quad (71)$$

where $\Sigma_{IJ} = \frac{1}{2} \epsilon_{IJKL} e^K \wedge e^L$ and F^{IJ} is the curvature of the Lorentz connection A^{IJ} . Since all backgrounds considered here have $R = R_0$, the field $\phi = f_R(R_0)$ is constant. Therefore, the connection equation $D(\phi \Sigma_{IJ}) = 0$ reduces to $D\Sigma_{IJ} = 0$, and no conformal rescaling of the tetrad is required [89].

We recall the relevant definitions [69–72, 90]. A co-dimension-one null surface Δ equipped with a degenerate metric q_{ab} of signature $(0, +, +)$ and an equivalence class of null normals $[C\ell^a]$, with C a positive constant, is a *non-expanding horizon* (NEH) if (i) it is topologically $S^2 \times \mathbb{R}$, (ii) its null normals are expansion free, $\theta_{(\ell)} \doteq 0$, (iii) $-T^a{}_b \ell^b$ is a future-directed causal vector field, and (iv) the field equations hold on Δ . An NEH is a *weak isolated horizon* (WIH) if it admits an equivalence class for which the connection on the normal bundle is Lie dragged along the generators, $\mathcal{L}_\ell \omega_{(\ell)} \doteq 0$. This characterization is quasi-local, since it describes a black hole in local equilibrium without requiring the exterior spacetime to be stationary or the existence of a global Killing field. The WIH boundary conditions break the bulk $SO(3, 1)$ symmetry to the subgroup preserving $[\ell^a]$, while boosts in the (ℓ, n) plane remain genuine horizon symmetries [91].

The on-shell variation of Eq. (71) is a pure boundary term, $16\pi G \delta \mathcal{A} = \int_{\partial M} \phi \Sigma_{IJ} \wedge \delta A^{IJ}$, from which the

symplectic potential and current follow as

$$16\pi G \theta(\delta) = \phi \Sigma_{IJ} \wedge \delta A^{IJ} , \quad (72)$$

$$16\pi G J(\delta_1, \delta_2) = \delta_1(\phi \Sigma_{IJ}) \wedge \delta_2 A^{IJ} - (1 \leftrightarrow 2) . \quad (73)$$

For a manifold without boundary, the closure of J guarantees conservation of the symplectic structure. An inner boundary modifies this result. However, since a WIH represents equilibrium, it must not act as a genuine flux source, and the pullback of J to Δ is exact, $J = dj$. Therefore, the horizon contribution reduces to the initial and final cross-sections, and the conserved symplectic structure acquires the boundary term

$$16\pi G \Omega(\delta_1, \delta_2) = \int_M \delta_1(\phi \Sigma_{IJ}) \wedge \delta_2 A^{IJ} - (1 \leftrightarrow 2) - \oint_{S_\Delta} \delta_1(\phi^2 \epsilon) \delta_2 \psi - \delta_2(\phi^2 \epsilon) \delta_1 \psi , \quad (74)$$

where ${}^2\epsilon$ is the volume form on the horizon cross sections and $\mathcal{L}_\ell \psi \hat{=} \kappa_{(\ell)}$, with $\psi = 0$ on the initial cross section of the horizon. The Hamiltonian function generating the residual boost labelled by the parameter η is then [92–94]:

$$H_\eta = \mathcal{N} \oint_{S_\Delta} \phi^2 \epsilon = \mathcal{N} f_R(R_0) A_\Delta , \quad (75)$$

where $\mathcal{N}$ is a normalization constant fixed by the Einstein limit, namely $\mathcal{N} = 1/(4G)$, or $\mathcal{N} = 1/4$ in units $G = 1$.

We now apply this construction to the three massive-branch reconstructions.

Rényi case. From Eq. (47),

$$\phi(R) = \frac{1}{2} e^{-\frac{3}{2}\pi c \lambda(1+cR)} [2 - 3\pi c \lambda(1+cR)] , \quad (76)$$

which can be inverted through the Lambert function as $R(\phi) = [2 - 3\pi c \lambda - 2W(e\phi)]/(3\pi c^2 \lambda)$. The Legendre transform (70) then gives

$$V(\phi) = \frac{3\pi c \lambda(2c\Lambda - \phi + 1) - 2\phi W(e\phi) - \frac{2\phi}{W(e\phi)} + 4\phi}{3\pi c^2 \lambda} . \quad (77)$$

Barrow case. From Eq. (49),

$$\phi(R) = \frac{\Delta + 2}{4} (\Delta \ln(6\pi c) - \Delta + 2) (1 + cR)^{\Delta/2} , \quad (78)$$

which can be inverted algebraically, yielding

$$V(\phi) = \frac{1}{c} \left[1 + 2c\Lambda - \phi + \frac{\Delta 16^{1/\Delta} \phi}{\Delta + 2} \times \left(\frac{\phi}{(\Delta + 2)(2 - \Delta + \Delta \ln(6\pi c))} \right)^{2/\Delta} \right] . \quad (79)$$

Kaniadakis case. From Eq. (51), $\phi(R) = \cosh[2\sqrt{3}\pi c \kappa(1 + cR)]$, and hence

$$R(\phi) = \frac{\cosh^{-1} \phi - 2\sqrt{3}\pi c \kappa}{2\sqrt{3}\pi c^2 \kappa} . \quad (80)$$

The corresponding potential is

$$V(\phi) = \frac{1}{6\pi c^2 \kappa} \left[6\pi c \kappa(2c\Lambda - \phi + 1) - \sqrt{3(\phi^2 - 1)} + \sqrt{3} \phi \cosh^{-1} \phi \right] . \quad (81)$$

Substituting each $\phi = f_R(R_0)$ into Eq. (75) and working in units $G = 1$, such that $\mathcal{N} = 1/4$, we obtain

$$H_\eta^{(\mathcal{R})} = \frac{A_\Delta}{4} \left[1 - \frac{\lambda A_\Delta}{8} + \mathcal{O}(\lambda^2) \right] , \quad (82)$$

$$H_\eta^{(B)} = \frac{A_\Delta}{4} \left[1 + \frac{\Delta}{2} \ln \left(\frac{A_\Delta}{4} \right) + \mathcal{O}(\Delta^2) \right] , \quad (83)$$

$$H_\eta^{(K)} = \frac{A_\Delta}{4} \left[1 + \frac{\kappa^2 A_\Delta^2}{96} + \mathcal{O}(\kappa^4) \right] . \quad (84)$$

These expressions coincide term by term with the perturbative Rényi, Barrow and Kaniadakis entropies. Hence, the generalized entropy is recovered not only as a thermodynamic input, but also as a conserved geometric charge associated with a horizon symmetry. Moreover, the same normalization $\mathcal{N} = 1/4$ applies in all three cases, without introducing any model-dependent rescaling.

V. CONCLUSIONS

Generalized horizon entropies provide a natural way to incorporate possible departures from the Bekenstein-Hawking area law and have been widely used in black-hole thermodynamics and cosmology. However, if the entropy of a horizon is modified, it is important to determine whether this modification can be associated with an underlying gravitational dynamics. In the present work we have addressed this inverse problem within metric $f(R)$ gravity, using the Wald entropy relation to reconstruct the gravitational Lagrangian from a prescribed entropy-area functional.

A central point of the analysis is that the entropy functional alone does not uniquely determine the Lagrangian. The Wald relation fixes f_R as a function of the horizon area, while the gravitational action requires f_R as a function of the curvature. Hence, an area-curvature relation $A(R)$ must first be specified. For Schwarzschild-de Sitter geometries the horizon area is not uniquely determined by the curvature, since the mass provides an additional parameter. Therefore, the reconstruction is defined only after selecting a one-parameter branch in the space of constant-curvature geometries. We have considered two physically distinct choices, namely the fixed-mass massive branch and the maximally symmetric branch.

The maximally symmetric branch is particularly useful, since its cosmological horizon obeys the exact and mass-independent relation $A = 48\pi/R$. As a result, the reconstruction reduces to the single quadrature (13), without any expansion in the entropy deformation parameter. We have applied this relation to the Bekenstein-Hawking, Tsallis-Cirto, Barrow, Rényi, Kaniadakis, logarithmically corrected and power-law entanglement entropies. The corresponding $f(R)$ Lagrangians are obtained in closed form and are summarized in Table I. In particular, the Bekenstein-Hawking entropy reproduces Einstein gravity identically, Tsallis-Cirto and Barrow entropies lead to power-law Lagrangians, Rényi entropy gives an elementary logarithmic structure, Kaniadakis entropy leads to a hyperbolic sine integral, the logarithmic entropy produces an $R^2 \ln R$ correction, and the power-law entanglement entropy generates an $R^{1-\beta}$ modification.

The exact construction also allows for several general results that go beyond the individual entropy models. An entropy contribution proportional to S_{BH}^q generates a curvature contribution proportional to R^{2-q} , with the exceptional case $q = 2$ giving a logarithmic term. Thus, on the maximally symmetric branch the entropy exponent is reflected about the extensive value $q = 1$. This mapping determines the curvature regime in which a given correction becomes important. In particular, entropy terms with $q > 1$ generate corrections with curvature power smaller than unity and therefore have an infrared character. A characteristic example is provided by the Kaniadakis entropy, whose leading correction generates the $1/R$ model of Carroll, Duvvuri, Trodden and Turner [65], with its scale fixed directly by the entropy deformation parameter. On the other hand, a logarithmic entropy correction generates the $R^2 \ln R$ structure associated with higher-curvature effective descriptions.

A second general result concerns the stability of the reconstructed theories. We have derived the branch-independent relation $f_{RR} = \frac{ds}{dR} \frac{d}{ds} \left(\frac{S}{s} \right)$, which converts the Dolgov-Kawasaki condition into a simple monotonicity criterion for $S(s)/s$. The sign of ds/dR is opposite on the two branches, and therefore the corresponding stability conditions are reversed. On the maximally symmetric branch Dolgov-Kawasaki stability requires $d(S/s)/ds < 0$, while on the massive branch it requires $d(S/s)/ds > 0$. Moreover, on the maximally symmetric branch we obtained the additional relation $m_{\text{sc}}^2 = \frac{S'(s)}{3f_{RR}}$, which directly connects the scalaron mass to the entropy functional. Consequently, for a positive and monotonically increasing entropy, ghost freedom, Dolgov-Kawasaki stability and the absence of a tachyonic scalaron can all be formulated through simple properties of the entropy itself.

We have additionally studied the massive branch, which is relevant to a black hole of fixed mass. In this case the area-curvature relation is obtained from an expansion around the Schwarzschild solution, independently of the perturbative expansion in the entropy deformation

parameter used in the main analysis. We reconstructed the Rényi, Barrow and Kaniadakis Lagrangians, determined the corresponding constant-curvature vacua and examined their scalaron stability. The Barrow reconstruction is stable throughout its physical parameter range, while the Kaniadakis reconstruction is perturbatively stable for sufficiently small deformation. The Rényi case provides an instructive contrast. Although the resummed expression possesses a finite interval in which the scalaron is non-tachyonic, this interval lies entirely in the region $f_{RR} < 0$. Hence, no positive- λ interval simultaneously satisfies the Dolgov-Kawasaki and non-tachyonic scalaron conditions. This agrees with the branch-independent criterion and with the exact massive-branch quadrature presented in Appendix A.

The comparison of the two branches shows that they should not be regarded as different approximations to the same reconstructed theory. They correspond to different curves in the space of constant-curvature geometries and thus lead to different Lagrangians. On the massive branch entropy powers are preserved through the dependence on $1 + cR$, while on the maximally symmetric branch they are reflected according to $q \rightarrow 2 - q$. Their stability properties are correspondingly different. This branch dependence is therefore not a technical detail but an intrinsic part of the entropy-Lagrangian correspondence.

Finally, we have examined the reconstructed theories in first-order form and within the covariant phase space of weak isolated horizons. The conserved charge associated with the residual horizon boost reproduces, with the same Einstein normalization, the generalized entropy from which each theory was constructed. This provides a quasi-local consistency check that is independent of the stationary Killing-horizon formulation used in the Wald reconstruction and further supports the thermodynamic interpretation of the resulting gravitational theories.

There are several directions in which the present framework can be extended. The most immediate one is the inclusion of matter and the study of the scalaron on cosmological and astrophysical backgrounds, which is necessary for a complete assessment of local-gravity and matter-era stability constraints. It would also be interesting to investigate the cosmological dynamics of the exact Lagrangians obtained here, especially those containing infrared corrections that may affect late-time acceleration. On the gravitational side, the reconstruction can be generalized to charged and rotating horizons, other exact branches such as Nariai-type geometries, higher-dimensional spacetimes and higher-curvature theories, including Lovelock gravity. Such extensions may clarify to what extent the entropy-Lagrangian correspondence found here represents a more general connection between modified horizon thermodynamics and gravitational dynamics.

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Appendix A: Exact quadratures on the massive branch

The perturbative character of the massive-branch analysis in Sec. IV A originates from the area-curvature map (7), which is obtained as a first-order expansion of the horizon cubic around the Schwarzschild point. It does not originate from the integration of the master relation itself. In fact, within the adopted massive-branch map the quadrature can be performed exactly in the entropy deformation parameter. Writing $s = s_0(1 + cR)$, with $s_0 = A_0/4G$, such that $ds = s_0 c dR$, Eq. (3) integrates to

$$f(R) = \frac{1}{s_0 c} \int^{s_0(1+cR)} \frac{S(s')}{s'} ds' + c_1. \quad (\text{A1})$$

For the three entropies considered in the main text this gives

$$f_B(R) = \frac{2}{s_0 c (2 + \Delta)} s^{1+\frac{\Delta}{2}} + c_1, \quad (\text{A2})$$

$$f_K(R) = \frac{1}{s_0 c \kappa} \text{Shi}(\kappa s) + c_1, \quad (\text{A3})$$

$$f_{\mathcal{R}}(R) = -\frac{1}{s_0 c \lambda} \text{Li}_2(-\lambda s) + c_1, \quad (\text{A4})$$

where Li_2 is the dilogarithm, $\text{Li}_2(z) = -\int_0^z \ln(1-t)t^{-1}dt$, and c_1 is fixed so that the undeformed limit reproduces $f(R) = R - 2\Lambda$. These expressions involve no expansion

in Δ , κ , or λ and have been verified by direct differentiation against Eq. (3).

A first observation concerns the exponent mapping. Equations (A2)-(A4) confirm directly that the massive branch preserves the entropy exponents. In particular, the Barrow entropy produces a pure power of $s \propto 1 + cR$, while on the maximally symmetric branch the same entropy generates the reflected power $R^{1-\Delta/2}$ according to Eq. (17).

A second observation concerns the relation between these exact quadratures and the closed expressions used in the main text. Equations (49), (51), and (47) are resummed forms constructed from the corresponding leading deformation expansions. They reproduce Eqs. (A2)-(A4) at the orders retained in the perturbative treatment, namely $\mathcal{O}(\Delta)$, $\mathcal{O}(\kappa^2)$, and $\mathcal{O}(\lambda)$, respectively, but they should not be identified with the exact quadratures at finite deformation. In particular, the exact Rényi reconstruction is dilogarithmic and the exact Kaniadakis reconstruction involves the hyperbolic sine integral. Thus, the expressions used in the main text are the appropriate ones for the perturbative analysis, while Eqs. (A2)-(A4) display the full dependence on the deformation parameters within the approximate map (7).

Finally, the exact quadratures provide an independent check of the branch-independent stability relation (31). Differentiating Eqs. (A2)-(A4) twice gives

$$\partial_R^2 f_B = \frac{\Delta}{2} s_0 c s^{\frac{\Delta}{2}-1} > 0, \quad (\text{A5})$$

$$\partial_R^2 f_K = s_0 c \frac{\kappa s \cosh(\kappa s) - \sinh(\kappa s)}{\kappa s^2} > 0, \quad (\text{A6})$$

$$\partial_R^2 f_{\mathcal{R}} = s_0 c \frac{1}{\lambda s^2} \left[\frac{\lambda s}{1 + \lambda s} - \ln(1 + \lambda s) \right] < 0, \quad (\text{A7})$$

where the signs follow from $x \cosh x > \sinh x$ and $x/(1+x) < \ln(1+x)$ for $x > 0$. Therefore, within the massive-branch map, the super-extensive Barrow and Kaniadakis entropies satisfy the Dolgov-Kawasaki condition for all values of their deformation parameters, whereas the sub-extensive Rényi entropy violates it for every $\lambda > 0$. This is precisely the inverted stability pattern predicted by Eq. (31), now obtained without expanding in the entropy deformation parameters.

This result also clarifies the Rényi stability window found in Sec. IV B. The apparent recovery of positivity for $\lambda > \lambda_p$ arises from the resummed Lagrangian (47) and is not a property of the exact Rényi quadrature within the adopted massive-branch map.

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